

Assignment – 4

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PART – 2

This command shows all modules available in PowerShell:

```
PS C:\Users\Administrator> Get-Module -ListAvailable

Directory: C:\Program Files\WindowsPowerShell\Modules

ModuleType Version      Name                                ExportedCommands
-----
Script      1.0.1        Microsoft.PowerShell.Operation.V... {Get-OperationValidation, Invoke-OperationValidat...
Binary      1.0.0.1      PackageManagement                  {Find-Package, Get-Package, Get-PackageProvider, ...
Script      3.4.0        Pester                             {Describe, Context, It, Should...}
Script      1.0.0.1      PowerShellGet                      {Install-Module, Find-Module, Save-Module, Update...
Script      2.0.0        PSReadLine                        {Get-PSReadLineKeyHandler, Set-PSReadLineKeyHandl...

Directory: C:\WINDOWS\system32\WindowsPowerShell\v1.0\Modules

ModuleType Version      Name                                ExportedCommands
-----
Manifest    1.0.1.0      ActiveDirectory                   {Add-ADCentralAccessPolicyMember, Add-ADComputerS...
Manifest    1.0.0.0      ADDSDeployment                   {Add-ADDSDeployment, Get-ADDSDeployment, Install...
Manifest    1.0.0.0      AppBackgroundTask                {Disable-AppBackgroundTaskDiagnosticLog, Enable-A...
Manifest    2.0.0.0      AppLocker                        {Get-AppLockerFileInformation, Get-AppLockerPolic...
Manifest    1.0.0.0      AppvClient                       {Add-AppvClientConnectionGroup, Add-AppvClientPac...
Manifest    2.0.1.0      Appx                             {Add-AppxPackage, Get-AppxPackage, Get-AppxPackag...
Manifest    1.0          BestPractices                    {Get-BpaModel, Get-BpaResult, Invoke-BpaModel, Se...
Script      2.0.0.0      BitsTransfer                     {Add-BitsFile, Complete-BitsTransfer, Get-BitsTra...
Manifest    1.0.0.0      BranchCache                     {Add-BCDataCacheExtension, Clear-BCCache, Disable...
Manifest    1.0.0.0      CimCmdlets                      {Get-CimAssociatedInstance, Get-CimClass, Get-Cim...
Manifest    1.0          ConfigCI                        {Get-SystemDriver, New-CIPolicyRule, New-CIPolicy...
Manifest    1.0          ConfigDefender                  {Get-MpPreference, Set-MpPreference, Add-MpPrefer...
Manifest    1.0          ConfigDefenderPerformance       {New-MpPerformanceRecording, Get-MpPerformanceRep...
Manifest    1.0          Defender                        {Get-MpPreference, Set-MpPreference, Add-MpPrefer...
Manifest    1.0          DefenderPerformance            {New-MpPerformanceRecording, Get-MpPerformanceRep...

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```

This command shows all commands inside the ServerManager module.

```
PS C:\Users\Administrator> Get-Command -Module ServerManager

CommandType      Name                                Version      Source
-----
Alias            Add-WindowsFeature                2.0.0.0      ServerManager
Alias            Remove-WindowsFeature              2.0.0.0      ServerManager
Function         Disable-ServerManagerStandardUserRemoting 2.0.0.0      ServerManager
Function         Enable-ServerManagerStandardUserRemoting 2.0.0.0      ServerManager
Cmdlet           Get-WindowsFeature                 2.0.0.0      ServerManager
Cmdlet           Install-WindowsFeature              2.0.0.0      ServerManager
Cmdlet           Uninstall-WindowsFeature            2.0.0.0      ServerManager

PS C:\Users\Administrator>
```

This command shows all features and roles installed or available.

```
PS C:\Users\Administrator> Get-WindowsFeature
```

Display Name	Name	Install State
[] Active Directory Certificate Services	AD-Certificate	Available
[] Certification Authority	ADCS-Cert-Authority	Available
[] Certificate Enrollment Policy Web Service	ADCS-Enroll-Web-Pol	Available
[] Certificate Enrollment Web Service	ADCS-Enroll-Web-Svc	Available
[] Certification Authority Web Enrollment	ADCS-Web-Enrollment	Available
[] Network Device Enrollment Service	ADCS-Device-Enrollment	Available
[] Online Responder	ADCS-Online-Cert	Available
[X] Active Directory Domain Services	AD-Domain-Services	Installed
[] Active Directory Federation Services	ADFS-Federation	Available
[] Active Directory Lightweight Directory Services	ADLDS	Available
[] Active Directory Rights Management Services	ADRMS	Available
[] Active Directory Rights Management Server	ADRMS-Server	Available
[] Identity Federation Support	ADRMS-Identity	Available
[] Device Health Attestation	DeviceHealthAttestat...	Available
[] DHCP Server	DHCP	Available
[X] DNS Server	DNS	Installed
[] Fax Server	Fax	Available
[X] File and Storage Services	FileAndStorage-Services	Installed
[X] File and iSCSI Services	File-Services	Installed
[X] File Server	FS-FileServer	Installed
[] BranchCache for Network Files	FS-BranchCache	Available
[] Data Deduplication	FS-Data-Deduplication	Available
[] DFS Namespaces	FS-DFS-Namespaces	Available
[] DFS Replication	FS-DFS-Replication	Available
[] File Server Resource Manager	FS-Resource-Manager	Available
[] File Server VSS Agent Service	FS-VSS-Agent	Available
[] iSCSI Target Server	FS-iSCSITarget-Server	Available
[] iSCSI Target Storage Provider (VDS and V...)	iSCSITarget-VSS-VDS	Available

This command filters and shows only installed features.

```
PS C:\Users\Administrator> Get-WindowsFeature | Where-Object {$_.InstallState -eq "Installed"}
```

Display Name	Name	Install State
[X] Active Directory Domain Services	AD-Domain-Services	Installed
[X] DNS Server	DNS	Installed
[X] File and Storage Services	FileAndStorage-Services	Installed
[X] File and iSCSI Services	File-Services	Installed
[X] File Server	FS-FileServer	Installed
[X] Storage Services	Storage-Services	Installed
[X] Web Server (IIS)	Web-Server	Installed
[X] Web Server	Web-WebServer	Installed
[X] Security	Web-Security	Installed
[X] Request Filtering	Web-Filtering	Installed
[X] Application Development	Web-App-Dev	Installed
[X] .NET Extensibility 3.5	Web-Net-Ext	Installed
[X] Management Tools	Web-Mgmt-Tools	Installed
[X] IIS Management Console	Web-Mgmt-Console	Installed
[X] .NET Framework 3.5 Features	NET-Framework-Features	Installed
[X] .NET Framework 3.5 (includes .NET 2.0 and 3.0)	NET-Framework-Core	Installed
[X] HTTP Activation	NET-HTTP-Activation	Installed
[X] Non-HTTP Activation	NET-Non-HTTP-Activ	Installed
[X] .NET Framework 4.8 Features	NET-Framework-45-Fea...	Installed
[X] .NET Framework 4.8	NET-Framework-45-Core	Installed
[X] ASP.NET 4.8	NET-Framework-45-ASPNET	Installed
[X] WCF Services	NET-WCF-Services45	Installed
[X] TCP Port Sharing	NET-WCF-TCP-PortShar...	Installed
[X] Group Policy Management	GPMC	Installed
[X] Microsoft Defender Antivirus	Windows-Defender	Installed
[X] Remote Server Administration Tools	RSAT	Installed
[X] Role Administration Tools	RSAT-Role-Tools	Installed
[X] AD DS and AD LDS Tools	RSAT-AD-Tools	Installed
[X] Active Directory module for Windows ...	RSAT-AD-PowerShell	Installed
[X] AD DS Tools	RSAT-ADDS	Installed

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This command shows information about Telnet Client feature:

```
PS C:\Users\Administrator> Get-WindowsFeature -Name Telnet-Client

Display Name                                     Name                                     Install State
-----
[ ] Telnet Client                               Telnet-Client                             Available

PS C:\Users\Administrator>
```

This command shows all system errors:

```
PS C:\Users\Administrator> Get-EventLog System -EntryType Error

Index Time                EntryType Source                                InstanceID Message
-----
4581 Apr 07 20:54 Error NETLOGON 5775 The dynamic deletion of the DNS record...
4580 Apr 07 20:54 Error NETLOGON 5775 The dynamic deletion of the DNS record...
4579 Apr 07 20:54 Error NETLOGON 5775 The dynamic deletion of the DNS record...
4578 Apr 07 20:54 Error NETLOGON 5775 The dynamic deletion of the DNS record...
4394 Apr 07 20:29 Error KDC 3221225479 The Security Account Manager failed a ...
4242 Apr 07 20:28 Error EventLog 2147489656 The previous system shutdown at 6:25:0...
4228 Apr 06 18:25 Error Service Control M... 3221232472 The Active Directory Web Services serv...
4227 Apr 06 18:25 Error Service Control M... 3221232481 A timeout was reached (30000 milliseco...
4221 Apr 06 18:23 Error Service Control M... 3221232472 The Active Directory Web Services serv...
4220 Apr 06 18:23 Error Service Control M... 3221232481 A timeout was reached (30000 milliseco...
4215 Apr 06 18:23 Error Service Control M... 3221232472 The Net.Pipe Listener Adapter service ...
4214 Apr 06 18:23 Error Service Control M... 3221232481 A timeout was reached (30000 milliseco...
4194 Apr 06 18:22 Error NETLOGON 5774 The dynamic registration of the DNS re...
4186 Apr 06 18:22 Error Service Control M... 3221232472 The MDCoreSvc service failed to start ...
4185 Apr 06 18:22 Error Service Control M... 3221232481 A timeout was reached (45000 milliseco...
4184 Apr 06 18:22 Error Service Control M... 3221232472 The WinDefend service failed to start ...
4183 Apr 06 18:22 Error Service Control M... 3221232481 A timeout was reached (45000 milliseco...
4180 Apr 06 18:22 Error Service Control M... 3221232472 The ADWS service failed to start due t...
4179 Apr 06 18:22 Error Service Control M... 3221232481 A timeout was reached (45000 milliseco...
4178 Apr 06 18:22 Error Service Control M... 3221232473 The NetTcpActivator service depends on...
4177 Apr 06 18:22 Error Service Control M... 3221232472 The NetTcpPortSharing service failed t...
4176 Apr 06 18:22 Error Service Control M... 3221232481 A timeout was reached (45000 milliseco...
4043 Apr 06 18:20 Error EventLog 2147489656 The previous system shutdown at 6:26:4...
3886 Apr 02 18:08 Error KDC 3221225479 The Security Account Manager failed a ...
3722 Apr 02 18:08 Error EventLog 2147489656 The previous system shutdown at 10:54:...
3631 Apr 01 22:10 Error KDC 3221225479 The Security Account Manager failed a ...
3448 Apr 01 21:24 Error KDC 3221225479 The Security Account Manager failed a ...
3284 Apr 01 21:24 Error EventLog 2147489656 The previous system shutdown at 8:42:0...
3161 Mar 29 20:26 Error KDC 3221225479 The Security Account Manager failed a ...
2998 Mar 29 20:26 Error EventLog 2147489656 The previous system shutdown at 7:00:3...
```

This shows the latest 3 warning events.

```
PS C:\Users\Administrator> Get-EventLog System -EntryType Warning -Newest 3

Index Time                EntryType Source                                InstanceID Message
-----
4587 Apr 07 20:54 Warning Microsoft-Windows... 12 Time Provider NtpClient: This machine ...
4573 Apr 07 20:54 Warning NETLOGON 5782 Dynamic registration or deregistration...
4550 Apr 07 20:49 Warning User32 2147484724 The reason supplied by user WS\Adminis...

PS C:\Users\Administrator>
```

This shows events from a specific source:

```
PS C:\Users\Administrator> Get-EventLog System -Source Microsoft-Windows-Kernel-General
```

Index	Time	EntryType	Source	InstanceID	Message
4568	Apr 07 20:52	Information	Microsoft-Windows...	15	The description for Event ID '15' in S...
4567	Apr 07 20:52	Information	Microsoft-Windows...	15	The description for Event ID '15' in S...
4557	Apr 07 20:49	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4556	Apr 07 20:49	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4555	Apr 07 20:49	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4547	Apr 07 20:49	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4546	Apr 07 20:49	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4544	Apr 07 20:49	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4541	Apr 07 20:49	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4540	Apr 07 20:49	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4506	Apr 07 20:39	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4493	Apr 07 20:39	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4424	Apr 07 20:29	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4423	Apr 07 20:29	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4421	Apr 07 20:29	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4413	Apr 07 20:29	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4411	Apr 07 20:29	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4407	Apr 07 20:29	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4382	Apr 07 20:29	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4265	Apr 07 20:28	Information	Microsoft-Windows...	24	The description for Event ID '24' in S...
4240	Apr 07 20:28	Information	Microsoft-Windows...	20	The description for Event ID '20' in S...
4233	Apr 07 20:28	Information	Microsoft-Windows...	25	The description for Event ID '25' in S...
4232	Apr 07 20:28	Information	Microsoft-Windows...	12	The description for Event ID '12' in S...
4231	Apr 06 18:25	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4135	Apr 06 18:20	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4100	Apr 06 18:20	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4099	Apr 06 18:20	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...
4092	Apr 06 18:20	Information	Microsoft-Windows...	16	The description for Event ID '16' in S...

This command shows all services on the system

```
PS C:\Users\Administrator> Get-Service
```

Status	Name	DisplayName
Running	ADWS	Active Directory Web Services
Stopped	ALG	Application Layer Gateway Service
Running	AppHostSvc	Application Host Helper Service
Stopped	AppIDSvc	Application Identity
Running	AppInfo	Application Information
Stopped	AppMgmt	Application Management
Stopped	AppReadiness	App Readiness
Stopped	AppVClient	Microsoft App-V Client
Running	AppXSvc	AppX Deployment Service (AppXSVC)
Stopped	aspnet_state	ASP.NET State Service
Running	AudioEndpointBu...	Windows Audio Endpoint Builder
Running	Audiosrv	Windows Audio
Stopped	AxInstSV	ActiveX Installer (AxInstSV)
Running	BFE	Base Filtering Engine
Running	BITS	Background Intelligent Transfer Ser...
Stopped	BluetoothUserSe...	Bluetooth User Support Service_33e162
Stopped	BluetoothUserSe...	Bluetooth User Support Service_60142
Running	BrokerInfrastru...	Background Tasks Infrastructure Ser...
Stopped	BTAGService	Bluetooth Audio Gateway Service
Stopped	BthAvctpSvc	AVCTP service
Stopped	bthserv	Bluetooth Support Service
Running	camsvc	Capability Access Manager Service
Stopped	CaptureService_...	CaptureService_33e162
Stopped	CaptureService_...	CaptureService_60142
Running	cbdhsvc_33e162	Clipboard User Service_33e162
Running	cbdhsvc_60142	Clipboard User Service_60142
Running	CDPSvc	Connected Devices Platform Service
Running	CDPUserSvc_33e162	Connected Devices Platform User Ser...
Running	CDPUserSvc_60142	Connected Devices Platform User Ser...

This shows details about the BITS service.

```
PS C:\Users\Administrator> Get-Service -Name BITS
```

Status	Name	DisplayName
Running	BITS	Background Intelligent Transfer Ser...

```
PS C:\Users\Administrator> |
```

This command starts the BITS service.

```
PS C:\Users\Administrator> Start-Service -Name BITS
PS C:\Users\Administrator> |
```

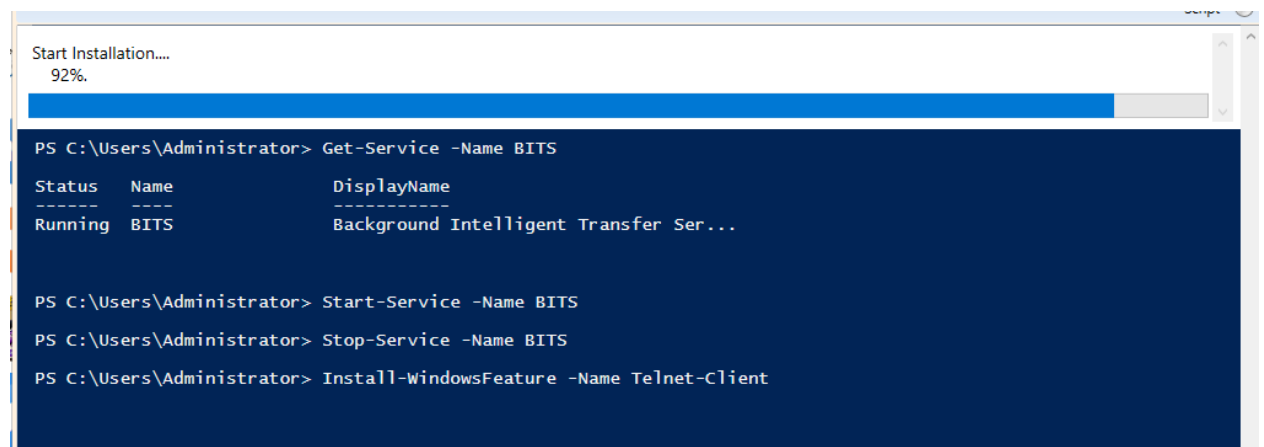
This command stops the BITS service.

```
PS C:\Users\Administrator> Stop-Service -Name BITS
PS C:\Users\Administrator>
```

Part 3

1.Install feature using the parameter Telnet-Client

This command installs Telnet Client feature:



```
PS C:\Users\Administrator> Get-Service -Name BITS

Status      Name      DisplayName
-----
Running     BITS      Background Intelligent Transfer Ser...

PS C:\Users\Administrator> Start-Service -Name BITS
PS C:\Users\Administrator> Stop-Service -Name BITS
PS C:\Users\Administrator> Install-WindowsFeature -Name Telnet-Client
```

```
PS C:\Users\Administrator> Install-WindowsFeature -Name Telnet-Client

Success Restart Needed Exit Code      Feature Result
-----
True      No                NoChangeNeeded {}
```

2.Show events from a system warnings which contain the message “*time service*”

This command shows warning events that contain "time service".

```
PS C:\Users\Administrator> Get-EventLog System -EntryType Warning | Where-Object {$_.Message -like "*time service*"}

Index Time      EntryType Source                                InstanceID Message
-----
4587 Apr 07 20:54 Warning Microsoft-Windows... 12 Time Provider NtpClient: This machine ...
4388 Apr 07 20:29 Warning Microsoft-Windows... 12 Time Provider NtpClient: This machine ...
4209 Apr 06 18:22 Warning Microsoft-Windows... 12 Time Provider NtpClient: This machine ...
3883 Apr 02 18:08 Warning Microsoft-Windows... 12 Time Provider NtpClient: This machine ...
3442 Apr 01 21:24 Warning Microsoft-Windows... 12 Time Provider NtpClient: This machine ...
3158 Mar 29 20:26 Warning Microsoft-Windows... 12 Time Provider NtpClient: This machine ...
2966 Mar 29 18:59 Warning Microsoft-Windows... 12 Time Provider NtpClient: This machine ...
2615 Mar 29 18:24 Warning Microsoft-Windows... 12 Time Provider NtpClient: This machine ...
2270 Mar 29 18:04 Warning Microsoft-Windows... 12 Time Provider NtpClient: This machine ...
1993 Mar 29 17:35 Warning Microsoft-Windows... 12 Time Provider NtpClient: This machine ...
209 Mar 20 01:40 Warning Microsoft-Windows... 52 The time service has set the time with...

PS C:\Users\Administrator>
```

3. Stop and then start service DPS

```
PS C:\Users\Administrator> Stop-Service -Name DPS  
PS C:\Users\Administrator> Start-Service -Name DPS  
PS C:\Users\Administrator> |
```

This command stops and then starts the Diagnostic Policy Service.

Conclusion:

In this assignment, I learned the basics of PowerShell and how to use it to manage a computer system. I practiced different commands to work with modules, services, and system events. I also learned how to filter data, get help for commands, and automate simple tasks.

By completing the tasks, I understood how PowerShell can make system administration easier and faster. It helps reduce manual work and allows users to control the system using commands. Overall, this assignment improved my understanding of PowerShell and its importance in IT and automation.